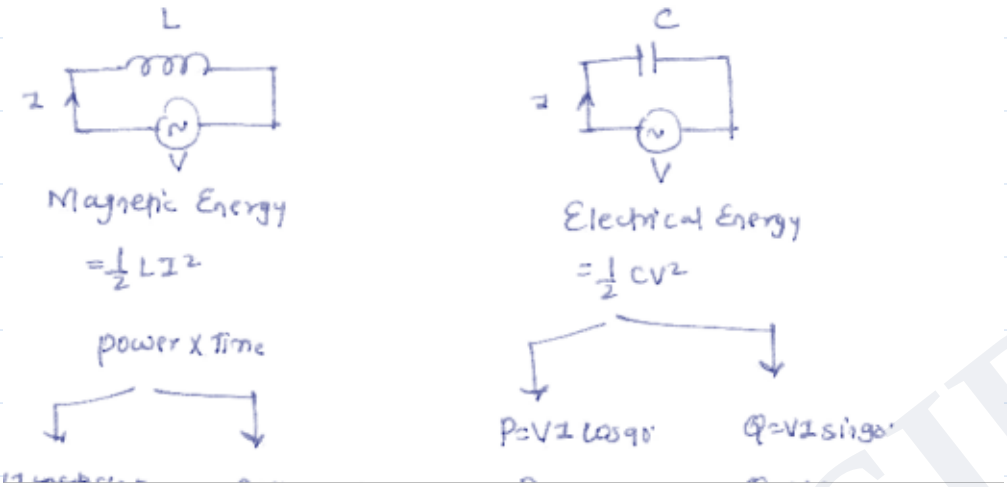
Voltage and Reactive Power Control

Class Notes

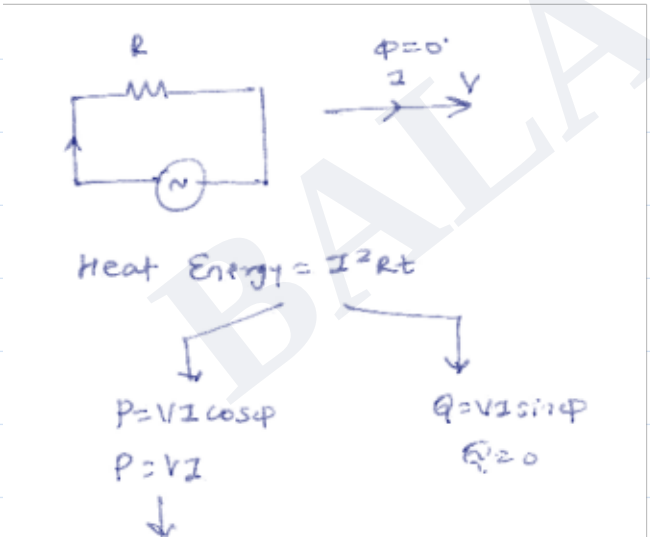
Notes by BALAJI SIR



Inductor L with source V, current I: magnetic energy = 1/2 L I^2

Magnetic energy = $\frac{1}{2}LI^2$; power x time : $Q = VI\sin\phi \Rightarrow Q = VI$; $P = VI\cos\phi \Rightarrow P = 0$

Electrical energy = $\frac{1}{2}CV^2$: $P = VI\cos 90^\circ \Rightarrow P = 0$; $Q = VI\sin 90^\circ = VI = \frac{1}{2}CV^2$



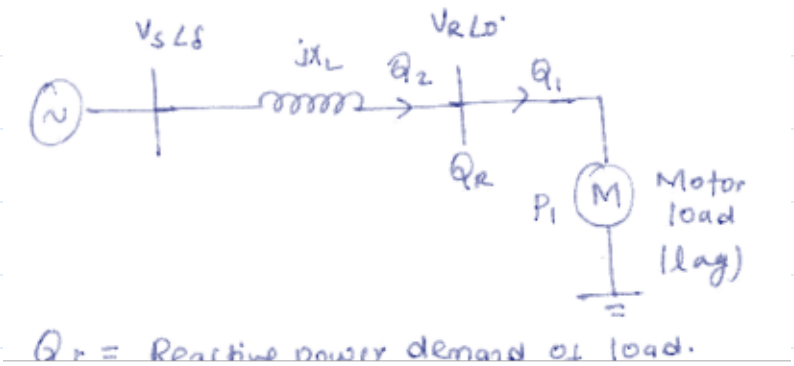
Resistor R with source: phi = 0 degrees

Heat energy = I^2Rt : $P = VI\cos\phi \Rightarrow P = VI \rightarrow I^2Rt$; $Q = VI\sin\phi \Rightarrow Q = 0$

Torque : $P = \omega T$, $T \propto P$

Relation between Vtg and Reactive power of transmission system :

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Vs at delta - jXL - VR at 0 degrees; Q2 (supply), Q1, QR; motor load P1 (lag)

Q_r = Reactive power demand of load ; Q_2 = Reactive power of supply

$$Q_R = Q_1 - Q_2$$

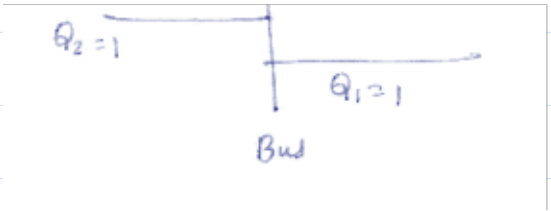
$$Q_R = \frac{V_S V_R}{X_L}\cos\delta - \frac{V_R^2}{X_L} \; ; \; \text{assume } \delta = 0 : \; Q_R = \frac{V_S V_R}{X_L} - \frac{V_R^2}{X_L}$$

$$V_R^2 - V_S V_R + X_L Q_R = 0 \quad (ax^2 + bx + c = 0)$$

$$V_R = \frac{V_S}{2} \pm \frac{\sqrt{V_S^2 - 4X_L Q_R}}{2} \Rightarrow V_R = \frac{V_S}{2} + \frac{\sqrt{V_S^2 - 4X_L Q_R}}{2}$$

Case 1 : $Q_R = 0 \Rightarrow Q_1 = Q_2$ (Demand = supply)

$V_R = V_S \Rightarrow V_{Reg} = 0$; V_R = const.

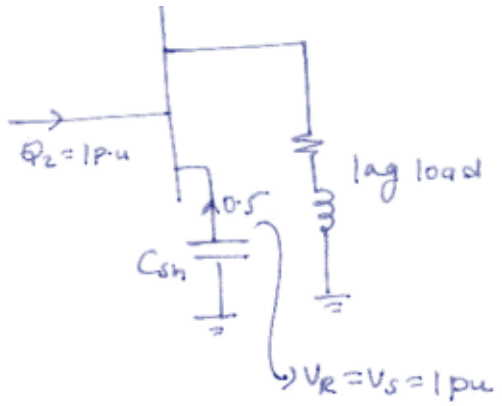


Bus with Q2 = 1 supply and Q1 = 1 demand

Case 2 : $Q_R > 0 \Rightarrow Q_1 > Q_2$ (Demand > Supply)

$V_R < V_S \Rightarrow V_{reg} > 0$; $V_S = 1$ p.u , $V_R = 0.8 < 1$

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Bus with supply Q2 = 1 p.u, shunt capacitor Csh supplying 0.5, lag load; VR = Vs = 1 p.u

Case 3 : $Q_R < 0 \Rightarrow Q_1 < Q_2$

$V_R > V_S$; $V_{reg} < 0$

Procedure for Designing Type and rating of compensator for maintaining ($V_R = V_S$)

$V_{reg} = 0$

(1) Assume UPF load {P_R is known}

(2) Calculate load angle δ by using : $P_R = \frac{V_S V_R}{X_L}\sin(\delta)$ -> for T/L with only X_L ; or

$P_R = \left|\frac{V_S V_R}{B}\right| \cos(\beta - \delta) - \left|\frac{AV_R^2}{B}\right| \cos(\beta - \alpha)$

(3) From the calculated value of (δ) calculate Q_R by using :

$$Q_{Rcal} = \left|\frac{V_S V_R}{X_L}\cos(\delta) - \frac{V_R^2}{X_L}\right| \text{ or } Q_{Rcal} = \left|\frac{V_R V_S}{B}\right| \sin(\beta - \delta) - \left|\frac{AV_R^2}{B}\right| \sin(\beta - \alpha)$$

(4) Calculate rating of shunt capacitor to be installed at the receiving end for supplying the reactive power to the Tr. line by using :

$Q_{cline} = -Q_{Rcal}$ = Rating of C_{sh} required to meet line reactive power demand from the receiving end.

(5) If the load is of lagging nature, calculate the reactive power requirement of the load by using {$Q_{Load} = P_R \cdot \tan\phi_R$} ; ϕ_R = p.f angle of load ; P_R = Active power consumption of load. Total reactive power requirement to both load and line :

$Q_{cTotal} = Q_{Load} + Q_{cLine}$ = Rating of C_{sh} (shunt capacitor)

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(6) For leading load calculate Q_{Load} equal to $P_R \cdot \tan\phi_R$: $Q_{Load} = P_R(\tan\phi_R)$

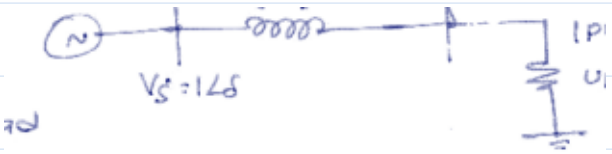
(a) $|Q_{Load}| > |Q_{cline}|$ then calculate shunt reactor rating of $Q_{Lsh} = |Q_{Load}| - |Q_{cLine}|$

(b) $|Q_{Load}| < |Q_{cline}|$ then shunt capacitor has to be installed with a rating of $Q_{Csh} = |Q_{cLine}| - |Q_{Load}|$

Eg: A Tr. line has reactance of 0.5 p.u is connected with a following load :

(i) 1 p.u UPF load. (ii) 1 p.u 0.8 lagging load. (iii) 1 p.u 0.6 leading load. (iv) 1 p.u 0.995 leading load.

Find the type and rating of compensator to be installed at the receiving end for maintaining $V_S = V_R = 1$ p.u



Vs = 1 at 0 degrees - j0.5 - VR = 1 at 0 degrees; UPF load 1 p.u = PR

(1) Assume UPF load

(2)

$$P_R = \left|\frac{V_S V_R}{X_L}\right| \sin\delta \; : \; 1 = \frac{1 \times 1}{0.5}\sin(\delta) \Rightarrow \delta = 30^\circ$$

(3)

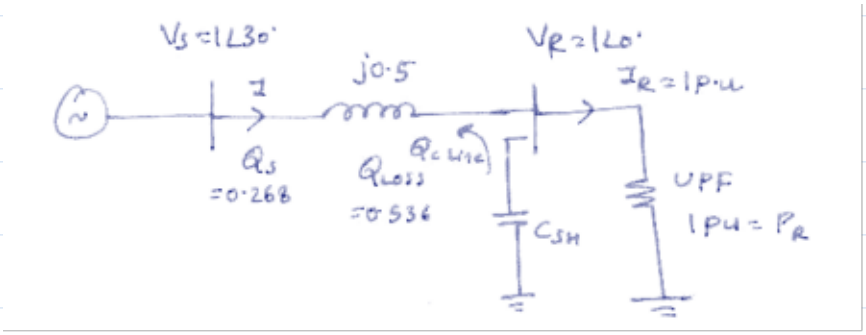
$$Q_{Rcal} = \left|\frac{V_S V_R}{X_L}\right| \cos\delta - \left|\frac{V_R^2}{X_L}\right| = \frac{1 \times 1}{0.5}\cos(30^\circ) - \frac{1^2}{0.5} = -0.268 \text{ p.u}$$

(4)

$$Q_{cLine} = -Q_{Rcal} = -(-0.268) = 0.268 = Q_{csh} = \text{Rating of } C_{sh}$$

M2 : (Analysis)

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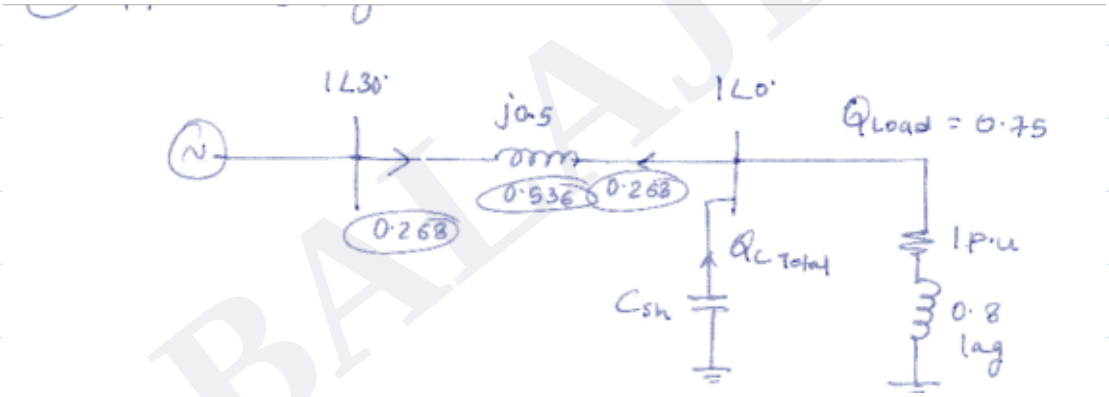
Vs = 1 at 30 degrees, j0.5, VR = 1 at 0 degrees, IR = 1 p.u; Qs = 0.268, Qloss = 0.536, Qc line, Csh; UPF load 1 p.u = PR

$$Q_{Loss} = |I^2 X_L| = \left(\frac{1\angle 30° - 1\angle 0°}{j0.5}\right)^2 (j0.5) = 0.536 \text{ p.u}$$

$$Q_s = \frac{V_s^2}{X_L} - \frac{V_S V_R}{X_L}\cos(\delta) = \frac{1^2}{0.5} - \frac{1\times 1}{0.5}\cos(30°) = 0.268 \text{ p.u}$$

$$Q_{c(line)} = 0.536 - 0.268 = 0.268 \text{ p.u } (Q_c \text{ rating of } C_{sh});\ Q_{Loss} = |2Q_s| = |2Q_R|$$

(2) 1 p.u 0.8 lag.



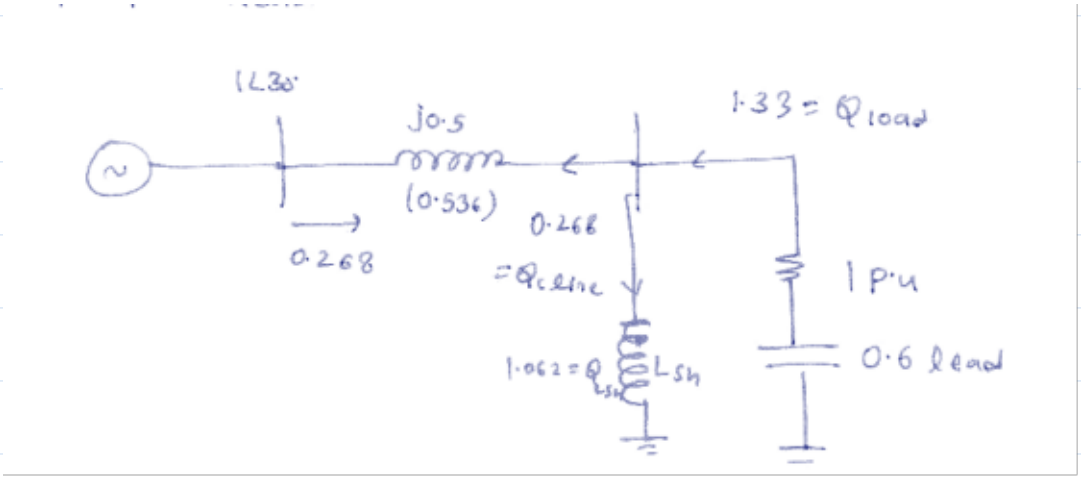
1 at 30 degrees, j0.5 (0.536, 0.268), 1 at 0 degrees; Qload = 0.75; Csh supplies Qc Total; load 1 p.u, 0.8 lag

$$Q_{Load} = P_R(\tan\phi_R) = 1\times\tan(\cos^{-1}0.8) = 0.75 \text{ p.u}$$

$$Q_{c\,Total} = Q_{c\,line} + Q_{c\,Load} = 0.268 + 0.75 = 1.018 \text{ p.u}$$

(3) 1 p.u, 0.6 lead.

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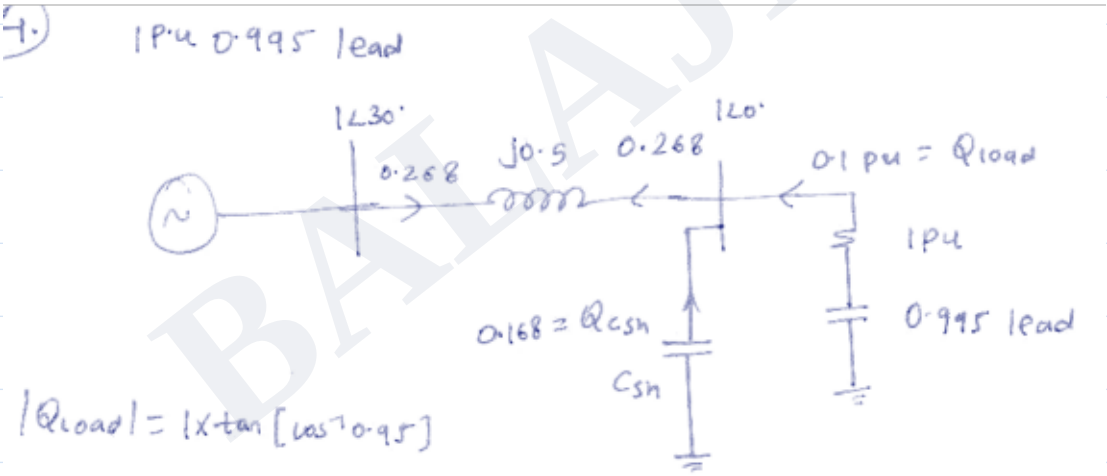
1 at 30 degrees, j0.5 (0.536), 1.33 = Qload, 0.268 = Qc line, Lsh absorbing 1.062; load 1 p.u, 0.6 lead

$$|Q_{Load}| = 1 \times \tan(\cos^{-1} 0.6) = 1.33, \quad |Q_{cline}| = 0.268$$

$Q_{load} > Q_{cline}$:

$$Q_{Lsh} = Q_{Load} - Q_{cline} = 1.33 - 0.268 = 1.062 \text{ p.u}$$

(4) 1 p.u 0.995 lead



1 at 30 degrees, j0.5 (0.268), 1 at 0 degrees; Qload = 0.1 p.u; Csh supplies 0.168; load 1 p.u, 0.995 lead

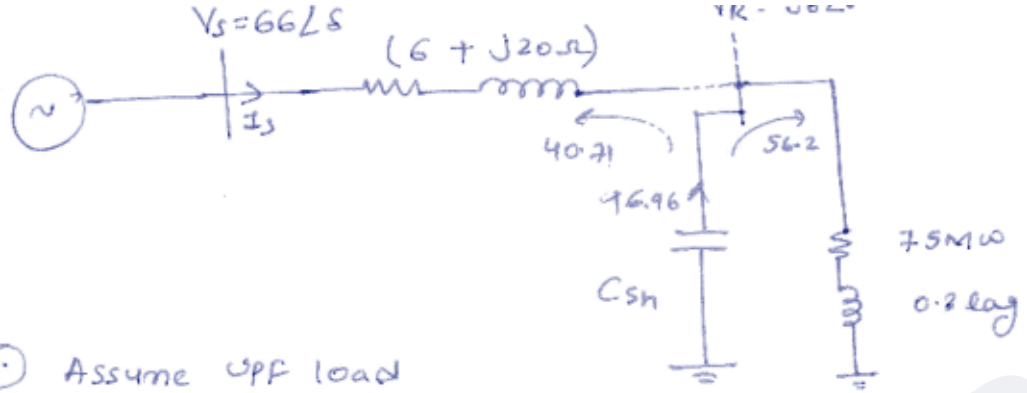
$$|Q_{Load}| = 1 \times \tan[\cos^{-1} 0.995] = 0.1, \quad Q_{cline} = 0.268 \text{ p.u}$$

$|Q_{load}| < |Q_{cline}|$:

$$Q_{Csh} = Q_{cline} - Q_{load} = 0.268 - 0.1 = 0.168 \text{ p.u}$$

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Eg: A 3-ϕ Tr. line has an impedance of $(6 + j20\,\Omega)$/ph connected with a load of 75 MW 0.8 lagging. Find the type and rating of the compensator to be installed at the receiving end for maintaining $V_S = V_R = 66$ kV



(1) Assume UPF load

Vs = 66 at delta - (6 + j20 ohm) - VR = 66 at 0 degrees; Csh 96.96 MVAR, line 40.71, load 75 MW 0.8 lag with 56.25

(1) Assume UPF load

(2) Calculate δ

$$\begin{bmatrix} A\angle\alpha & B\angle\beta \\ C\angle\Delta & D\angle\alpha \end{bmatrix} = \begin{bmatrix} 1\angle 0^\circ & (6+j20) = 20.88\angle 73.3^\circ \\ 0 & 1\angle 0^\circ \end{bmatrix}$$

$$P_R = \left|\frac{V_S V_R}{B}\right| \cos(\beta - \delta) - \left|\frac{AV_R^2}{B}\right| \cos(\beta - \alpha)$$

$$75 = \left|\frac{66^2}{20.88}\right| \cos(73.3 - \delta) - \left|\frac{1 \times 66^2}{20.88}\right| \cos(73.3 - 0) \Rightarrow \delta = 23.6^\circ$$

(3)

$$Q_{R\,cal} = \left|\frac{V_S V_R}{B}\right| \sin(\beta - \delta) - \left|\frac{AV_R^2}{B}\right| \sin(\beta - \alpha)$$

$$= \left|\frac{66^2}{20.88}\right| \sin(73.3 - 23.6) - \left|\frac{1 \times 66^2}{20.88}\right| \sin(73.3 - 0) = -40.71\ \text{MVAR}$$

(4)

$$Q_{c\,line} = -Q_{R\,cal} = +40.71$$

(5)

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$$Q_{Load} = P_R \tan\phi_R = 75 \times \tan(\cos^{-1} 0.8) = 56.25\ MVAR$$

(6)

$$Q_{CTOTAL} = Q_{cLoad} + Q_{cLine} = 40.71 + 56.25 = 96.96\ MVAR = C_{sh}$$

Condition for P_{max}

$$P_R = \left|\frac{V_S V_R}{B}\right| \cos(\beta - \delta) - \left|\frac{AV_R^2}{B}\right| \cos(\beta - \alpha)$$

δ is a variable ; for Max P_R => $\frac{dP_R}{d\delta} = 0$

$$\sin(\beta - \delta) = 0 \Rightarrow \delta = \beta$$

$$P_{Rmax} = \left|\frac{V_S V_R}{B}\right| - \left|\frac{AV_R^2}{B}\right| \cos(\beta - \alpha) \quad \text{At } \delta = \beta$$

$$Q_R = -\left|\frac{AV_R^2}{B}\right| \sin(\beta - \alpha) \quad \text{condition for } P_{max}$$

NOTE : Q_R is negative indicates leading reactive power has to be supplied for leading load (capacitor) for receiving of P_{max} active power.

Eg: In the previous problem find the Max Active power received at UPF and rating of compensator to be needed for receiving of Max Active power.

$$P_{Rmax} = \left|\frac{66^2}{20.88}\right| - \left|\frac{1 \times 66^2}{20.88}\right| \cos(73.3 - 0) = 148.68\ MW$$

$$Q_R = -\left|\frac{1 \times 66^2}{20.88}\right| \sin(73.3 - 0) = -199.82$$

$$Q_{Rcal} = -199.82 ; \quad Q_{csh} = -Q_{Rcal} = 199.82$$

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Locus of powers :

$$P_R = \left|\frac{V_S V_R}{B}\right| \cos(\beta - \delta) - \left|\frac{AV_R^2}{B}\right| \cos(\beta - \alpha)$$

$$P_R + \left|\frac{AV_R^2}{B}\right| \cos(\beta - \alpha) = \left|\frac{V_S V_R}{B}\right| \cos(\beta - \delta) \quad -(1)$$

$$Q_R + \left|\frac{AV_R^2}{B}\right| \sin(\beta - \alpha) = \left|\frac{V_S V_R}{B}\right| \sin(\beta - \delta) \quad -(2)$$

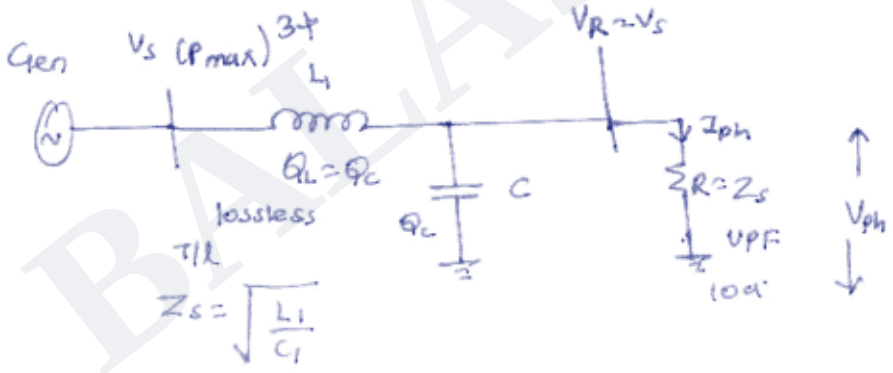
$(1)^2 + (2)^2$:

$$\left[P_R + \left|\frac{AV_R^2}{B}\right| \cos(\beta - \alpha)\right]^2 + \left[Q_R + \left|\frac{AV_R^2}{B}\right| \sin(\beta - \alpha)\right]^2 = \left(\frac{V_S V_R}{B}\right)^2 \quad -(3)$$

Eqn (3) is a circle $(x-a)^2 + (y-b)^2 = r^2$. Locus of (P_R, Q_R) (P_S, Q_S) is a circle.

DAY-6 [14/9/25]

Surge Impedance loading (ideal power T/F capability)



Gen Vs (Pmax) - 3-phase lossless T/L (L1, C1, Zs = sqrt(L1/C1); QL = QC) - VR = Vs - UPF load R = Zs, Iph, Vph

SIL = 3-ϕ Maximum Active power capability of Tr. line to UPF load

$$= 3V_{ph} I_{ph} \cos\phi = 3V_{ph} \cdot \frac{V_{ph}}{R} \cdot 1 = \frac{3V_{ph}^2}{Z_s}$$

$$SIL = 3\left(\frac{V_L}{\sqrt{3}}\right)^2 \times \frac{1}{Z_s}$$

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$$SIL = \frac{V_L^2}{Z_s} = V_L^2\sqrt{\frac{C_1}{L_1}} = V_s^2\sqrt{\frac{C_1}{L_1}}$$

$$SIL \propto V_s^2 \propto \sqrt{C_1} \propto \frac{1}{\sqrt{L_1}}$$

$L_1\downarrow$: Bundle conductor, C_{se}, parallel T/L

Effect of compensator on SIL :

Z_s = Surge Impedance without compensator = $\sqrt{\frac{L_1}{C_1}}$

Z_{sc} = Surge Impedance with compensator

(SIL) = (SIL) w/o compensator

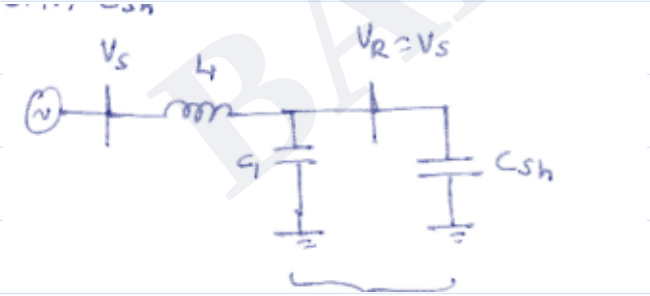
$(SIL)_c$ = SIL with compensator

K_{Csh} = % or Degree of C_{sh} compensator

K_{Lsh} = % or Degree of L_{sh} compensator

K_{se} ... C_{se}

With C_{sh}



Vs - L1 - C1 and Csh at the receiving end (VR = Vs)

$$C_T = C_1 + C_{sh} = C_1\left[1 + \frac{C_{sh}}{C_1}\right] = C_1[1 + K_{Csh}], \quad K_{Csh} = \frac{C_{sh}}{C_1}$$

$$Z_{sc} = \sqrt{\frac{L_1}{C_T}} = \sqrt{\frac{L_1}{C_1(1 + K_{Csh})}}$$

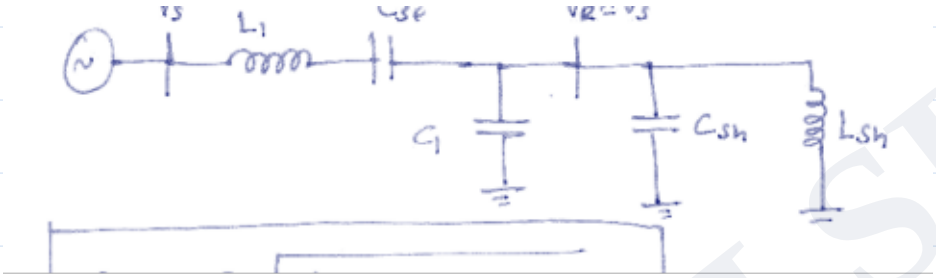
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$$Z_{sc} = \frac{Z_s}{\sqrt{(1+K_{Csh})}}$$

$$(SIL)_c = \frac{V_s^2}{Z_{sc}} = \frac{V_s^2}{Z_s}\sqrt{1+K_{sh}}$$

$$(SIL)_c = (SIL)\sqrt{1+K_{Csh}}$$

With C_{sh}, C_{se}, L_{sh}

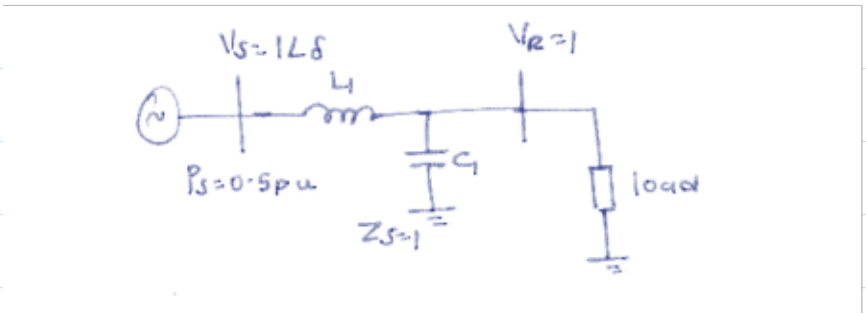


Vs - L1 - Cse - C1 - VR = Vs with Csh and Lsh at the receiving end

$$Z_{sc} = Z_s\sqrt{\frac{(1-K_{se})}{(1+K_{Csh})(1-K_{Lsh})}}$$

$$(SIL)_c = (SIL)\sqrt{\frac{(1+K_{Csh})(1-K_{Lsh})}{(1-K_{se})}}$$

W/o compensator :

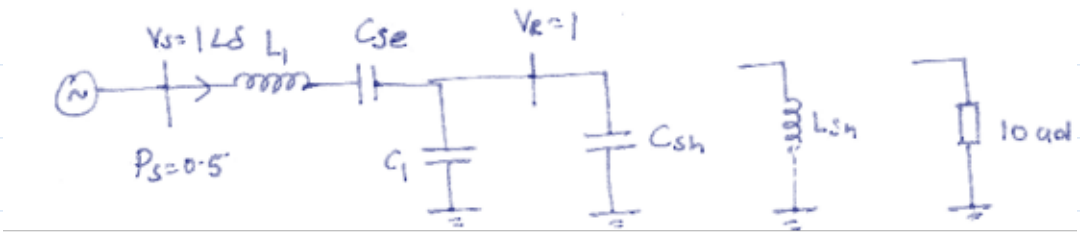


Vs = 1 at delta - L1 - C1 - VR = 1 - load; Ps = 0.5 p.u, Zs = 1

$$P_{max} = SIL = \left|\frac{V_s^2}{Z_s}\right| = \frac{1^2}{1} = 1\ p.u$$

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$$P_s = P_{max}\sin\delta \;:\; 0.5 = 1 \times \sin\delta \Rightarrow \delta = 30^\circ$$



Compensated line: Vs = 1 at delta, L1, Cse, C1, VR = 1, Csh, Lsh, load; Ps = 0.5

$K_{se} = K_{Csh} = K_{Lsh} = 30\% = 0.3$

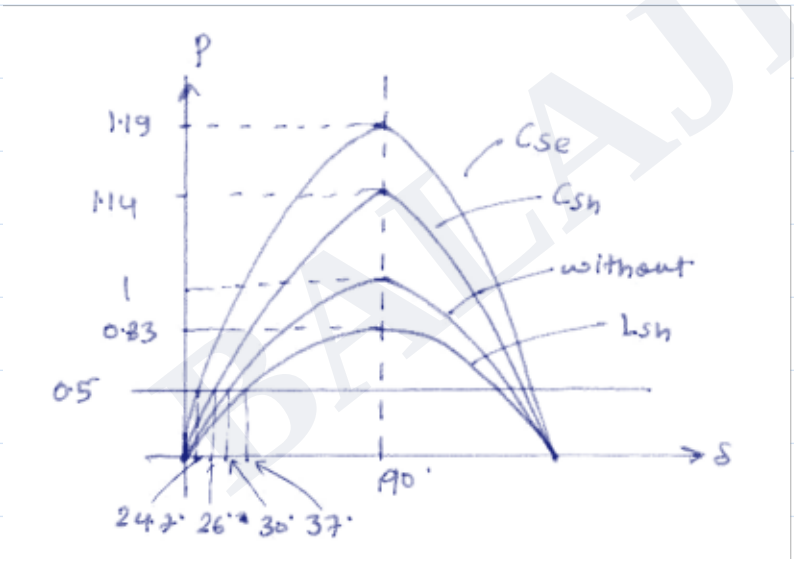
Component Z_{sc} $(SIL)_c = P_{max}$ $\delta = \sin^{-1}\left(\frac{P_s}{P_{max}}\right)$

w/o compensator 1 1 30°

$K_{Csh} = 30\%$ $Z_{sc} = \dfrac{1}{\sqrt{1+0.3}} = 0.87$ $1 \times \sqrt{1+0.3} = 1.14$ $\delta = \sin^{-1}\left(\dfrac{0.5}{1.14}\right) = 26^\circ$

$K_{se} = 30\%$ $Z_{sc} = 1 \times \sqrt{1-0.3} = 0.83$ $\dfrac{1}{\sqrt{1-0.3}} = 1.19$ $\delta = \sin^{-1}\left(\dfrac{0.5}{1.19}\right) = 24.7^\circ$

$K_{Lsh} = 30\%$ $Z_{sc} = \dfrac{1}{\sqrt{1-0.3}} = 1.19$ $\sqrt{1-0.3} = 0.83$ $\delta = \sin^{-1}\left(\dfrac{0.5}{0.83}\right) = 37^\circ$



P - delta curves: Cse (peak 1.19), Csh (1.14), without compensator (1), Lsh (0.83); Ps = 0.5 line cuts at 24.7, 26, 30 and 37 degrees

Power Transfer capability and stability: $C_{se} > C_{sh} >$ (w/o) $> L_{sh}$

NOTE :

MW loading = SIL => $V_R = V_S$ -> No compensation required

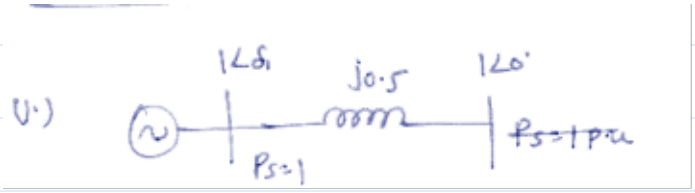
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Loading < SIL => $V_R > V_S$ => C_{sh} required for $V_R = V_S$ (off peak load loading time)

Loading > SIL => $V_R < V_S$ => C_{sh} required for $V_R = V_S$ (peak loading time)

Midpoint compensation :

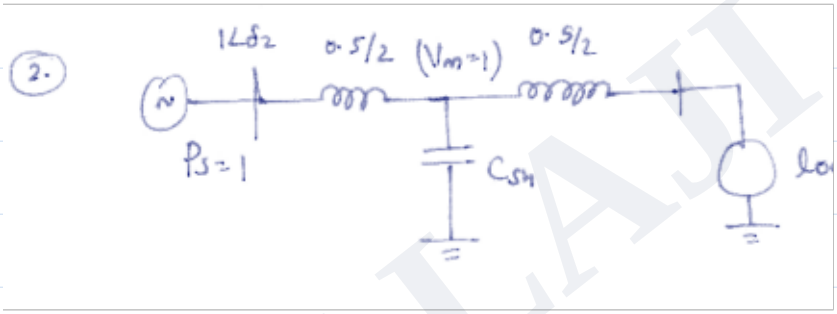
(1)



1 at delta1 - j0.5 - 1 at 0 degrees; Ps = 1 p.u

$$P_s = \frac{1 \times 1}{0.5} \sin(\delta_1) \Rightarrow \delta_1 = 30°$$

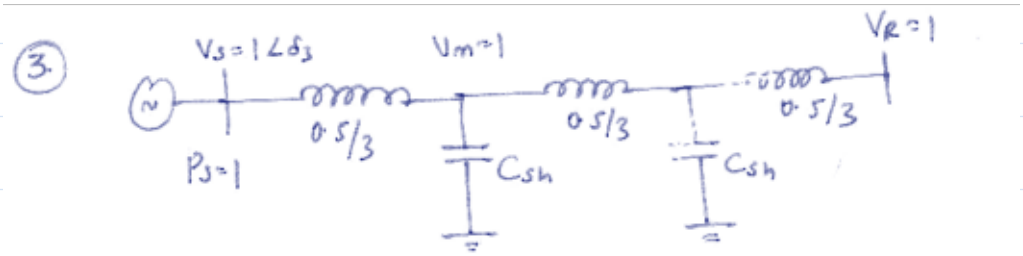
(2)



1 at delta2 - 0.5/2 - (Vm = 1, Csh) - 0.5/2 - load; Ps = 1

$$P_s = 1 = \left|\frac{1 \times 1}{0.5/2}\right| \sin(\delta_2) \quad \delta_2 = 14.7° \approx 15° \Rightarrow \frac{\delta_1}{2}$$

(3)



Vs = 1 at delta3 - 0.5/3 - Vm = 1, Csh - 0.5/3 - Csh - 0.5/3 - VR = 1

$$P_s = 1 = \left|\frac{1 \times 1}{0.5/3}\right| \sin\delta_3 \quad \delta_3 = 9.8° \approx 10° = \left(\frac{\delta_1}{3}\right)$$

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(4) For n-equal sections :



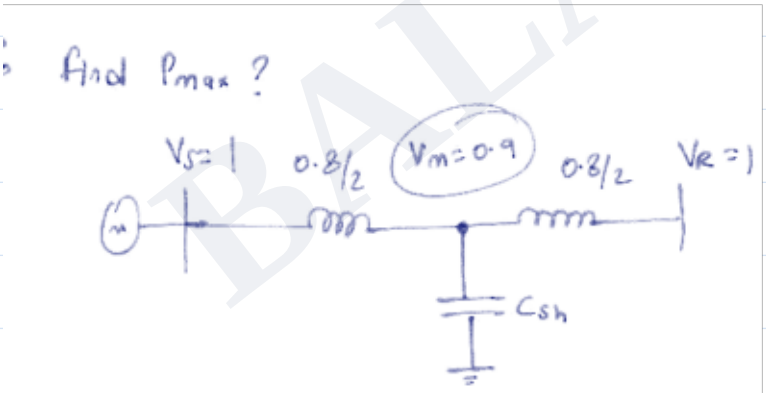
Vs at delta0 - XL/m - Vm, Csh - XL/m ... - Csh - XL/m - VR

$$P_s = \left|\frac{V_s V_m}{X_{L/m}}\right| \sin(\delta_0) \qquad V_s = V_m$$

$$\boxed{P_s = \left|\frac{m V_s V_m}{X_L}\right| \sin\left(\frac{\delta}{m}\right)}$$

$$P_s = \left|\frac{m V_s^2}{X_L}\right| \sin\left(\frac{\delta}{m}\right)$$

Eg: Find P_{max} ?

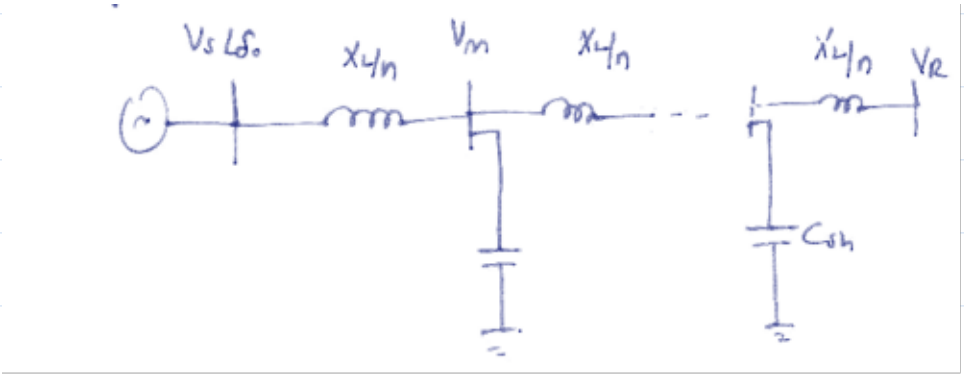


Vs = 1 - 0.8/2 - (Vm = 0.9, Csh) - 0.8/2 - VR = 1

$$P_{max} = \left|\frac{V_s V_m}{X_L}\right| = \left|\frac{1 \times 0.9}{0.8/2}\right| = \frac{9}{4} = 2.25 \text{ p.u}$$

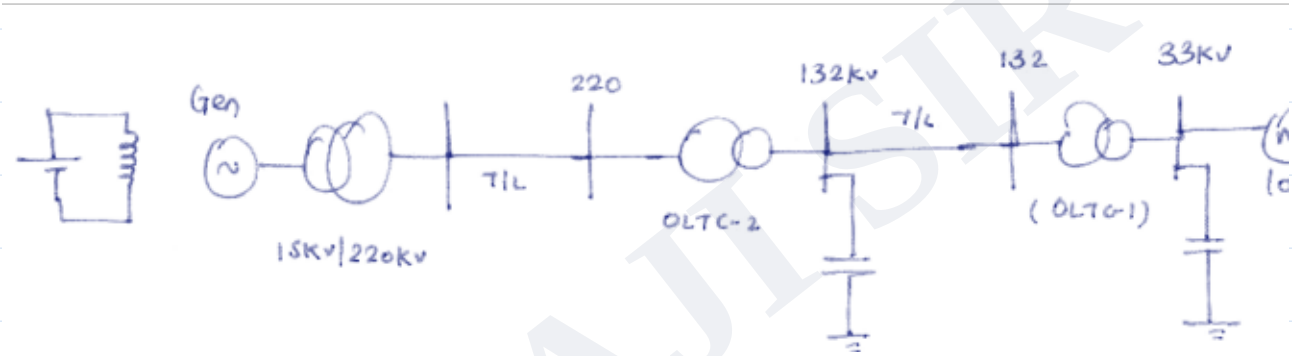
For n equal sections :

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Vs at delta0 - XL/n - Vm - XL/n ... - XL/n - VR, with Csh at each section

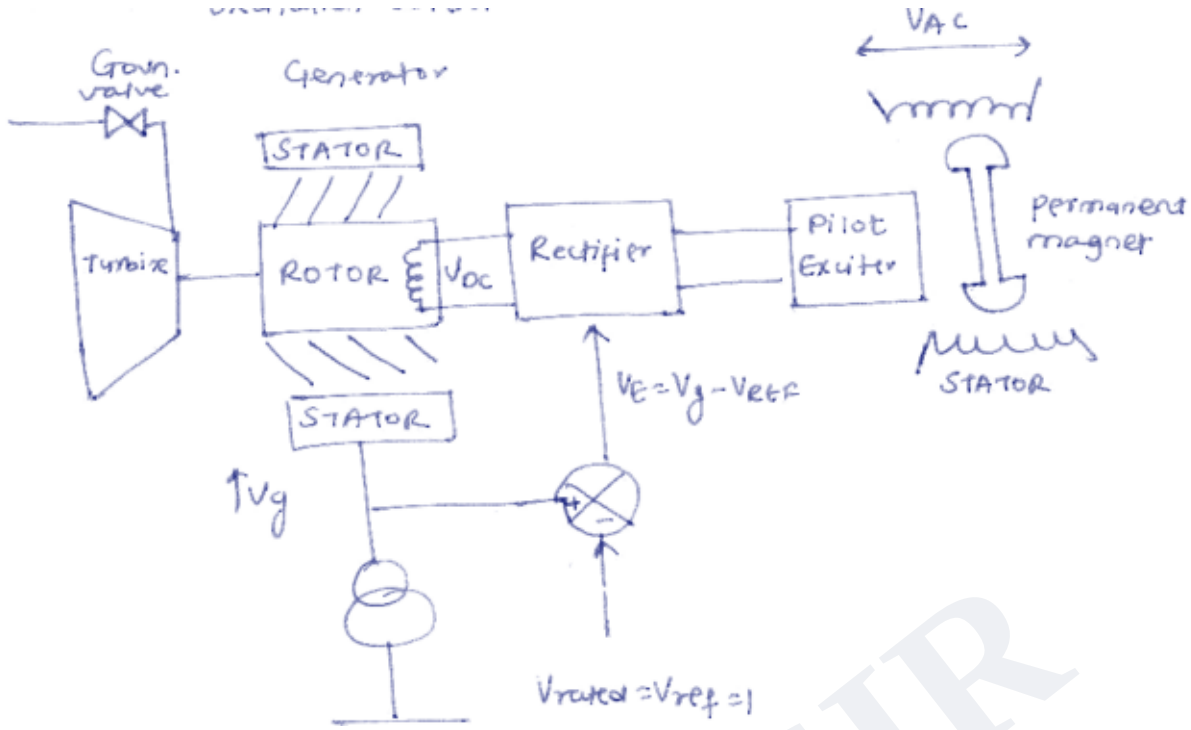
$$P_s = \left| \frac{V_s V_m}{X_L/n} \right| \sin(\delta_0)$$



Generator, 15 kV/220 kV transformer, T/L, OLTC-2 at 132 kV with shunt capacitor, T/L, OLTC-1 at 33 kV with shunt capacitor, motor load

Generator Excitation control :

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Turbine with governor valve, rotor field fed through rectifier from pilot exciter (permanent magnet); generator terminal voltage Vg compared with Vrated = Vref = 1 to give VE = Vg - Vref

$$V_E = V_g - V_{Rated}$$

(1) If $V_g = V_{Rated} \Rightarrow V_E = 0$ Const. Excitation

(2) If $V_g > V_{Rated} \Rightarrow V_E > 0$ Reduce excitation until $V_E = 0$

(3) If $V_g < V_{Rated} \Rightarrow V_E < 0$ Increase excitation until $V_E = 0$

Sequence for Vtg control: (1) C_{sh} ; (2) OLTC ; (3) Generator Excitation

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